Ramtin Moghaddam

Project Finished together with **Sidney Mills**

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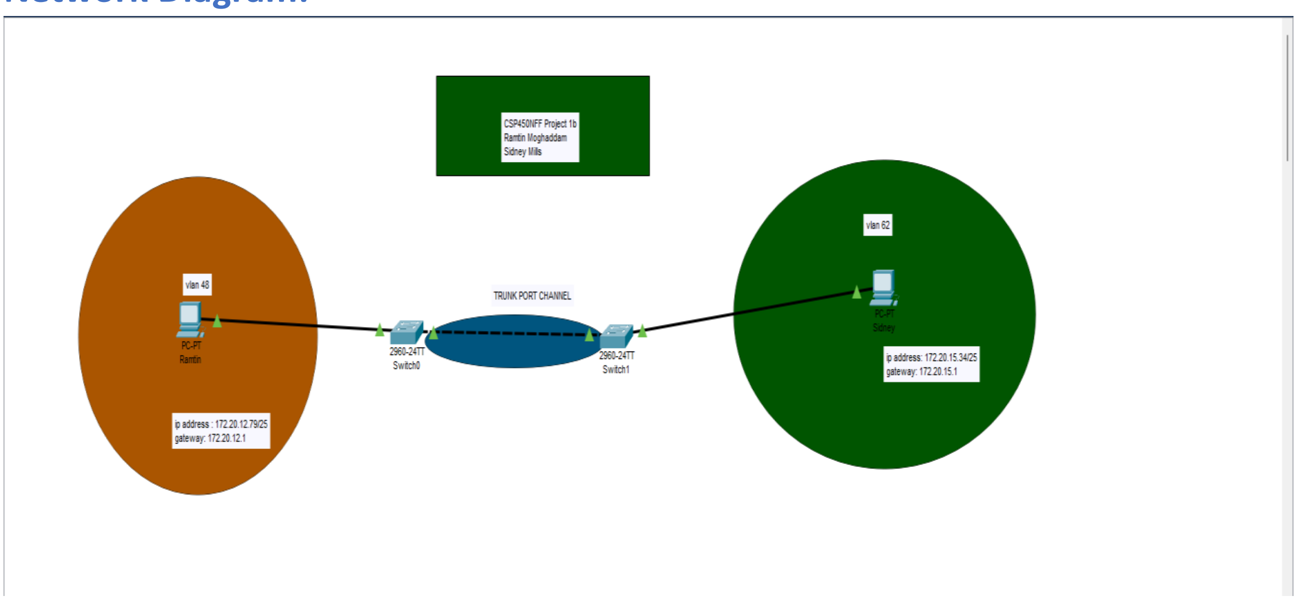
1. Network Diagram:



2. The script Used to configure the switch

2960 switch (Ramtin):

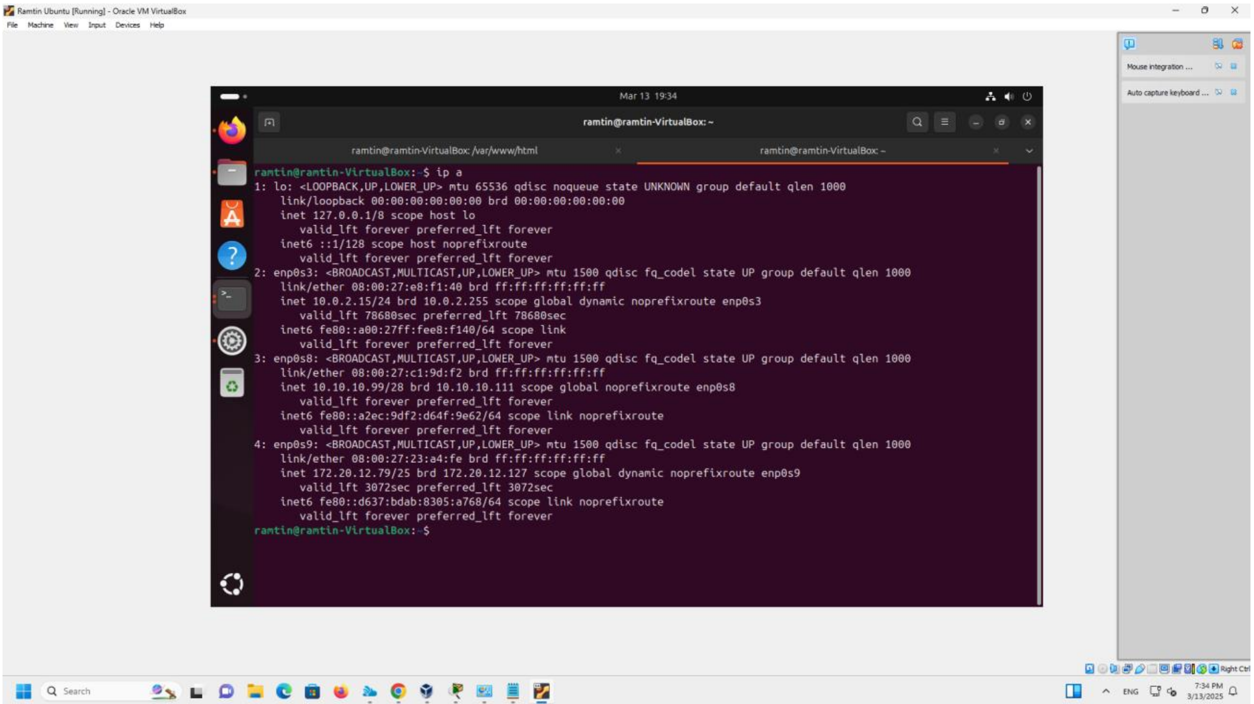
```
config t
Vlan 62
  Name vlan 62
  Exit
Vlan 48
  Name vlan 48
  Exit
Interface 1/1/5
  no shutdown
  no routing
  vlan trunk allowed 48,62
  exit
interface 1/1/6
  no shutdown
  no routing
  vlan trunk allowed 48,62
  exit
interface 1/1/1
  no shutdown
  no routing
  vlan access 48
  exit

int vlan 48
  ip address 172.20.12.1 255.255.255.128
  exit
int vlan 62
  ip address 172.20.15.1 255.255.255.128
int lag 1
  no shut
```

```
  no route
  vlan trunk allowed all
  lacp mode active
dhcp-server vrf default
  pool Ramtin
      range 172.20.12.2 172.20.12.126
      default-router 172.20.12.1
      exit
   pool Sidney
      range 172.20.15.2 172.20.15.126
      default-router 172.20.15.1
      exit
authoritative
enable
```

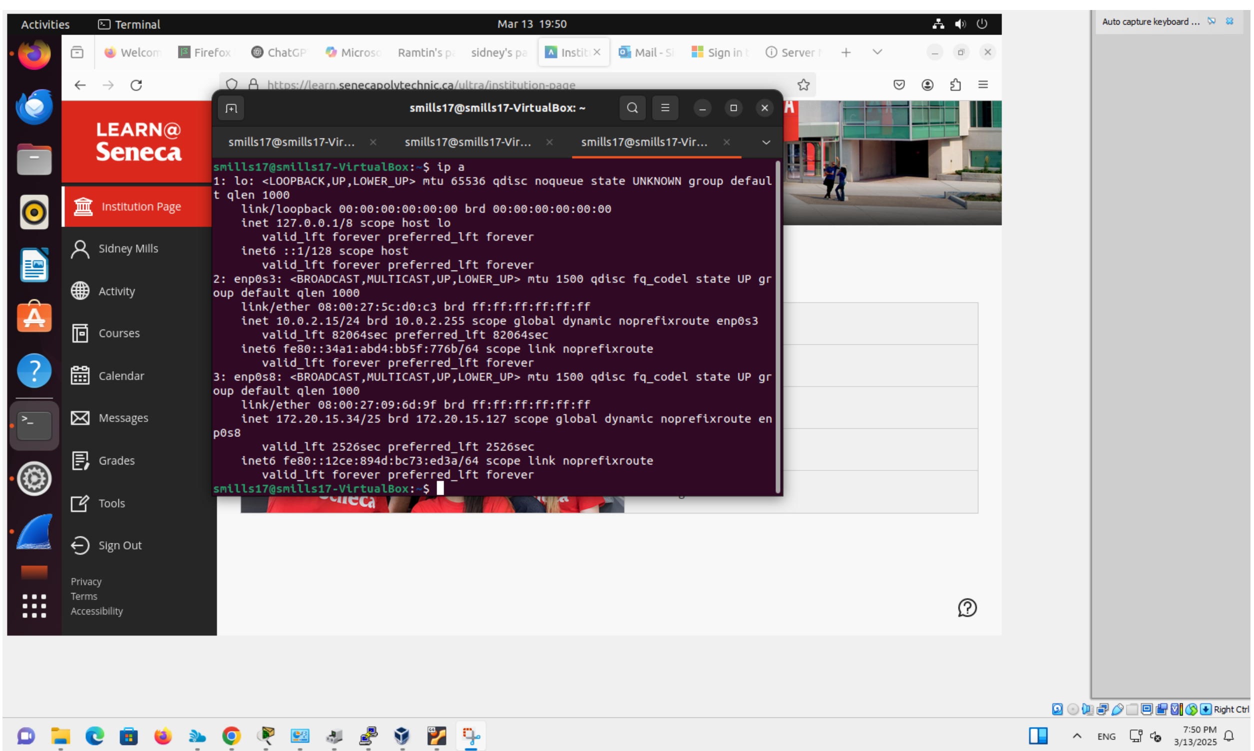
2530 switch (Sidney):

```
Vlan 1
  name "DEFAULT_VLAN"
  no untagged 1
  untagged 4-28, Trk1
  ip address 10.10.10.88 255.255.255.240
  exit
vlan 62
  name "VLAN62"
  untagged 1
  tagged Trk1
  no ip address
  exit
spanning-tree Trk1 priority 4
```

3. Ip a

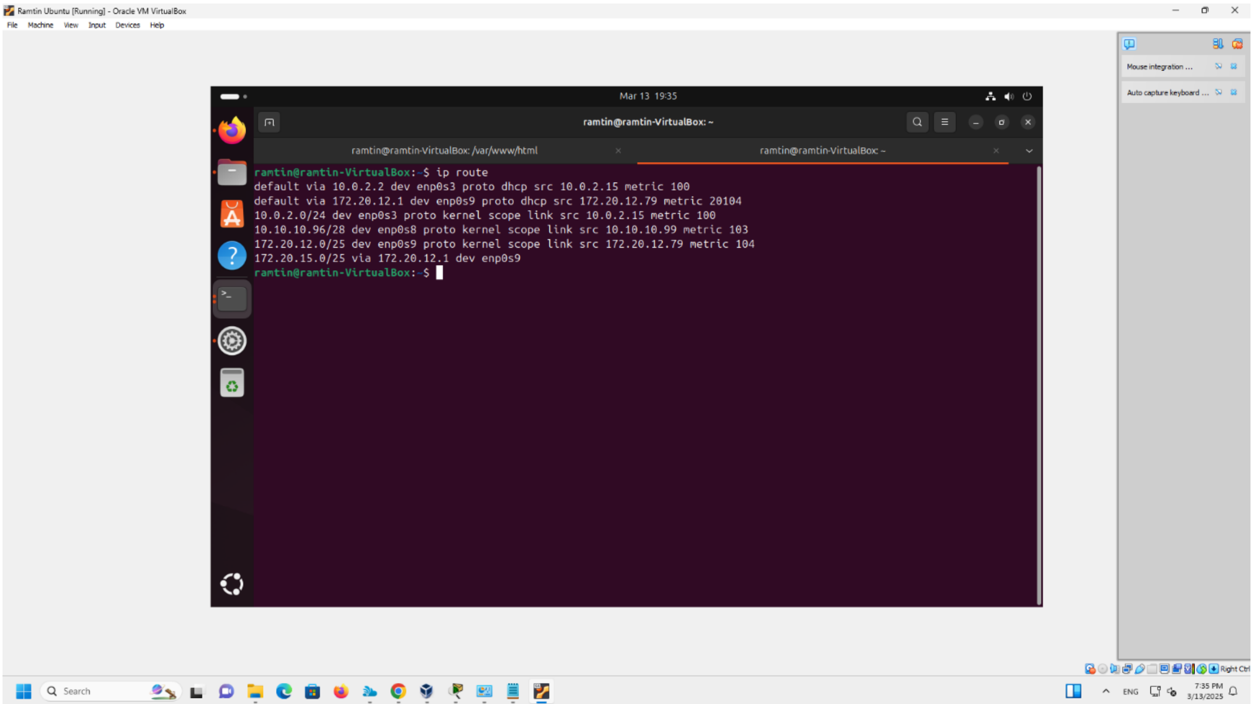


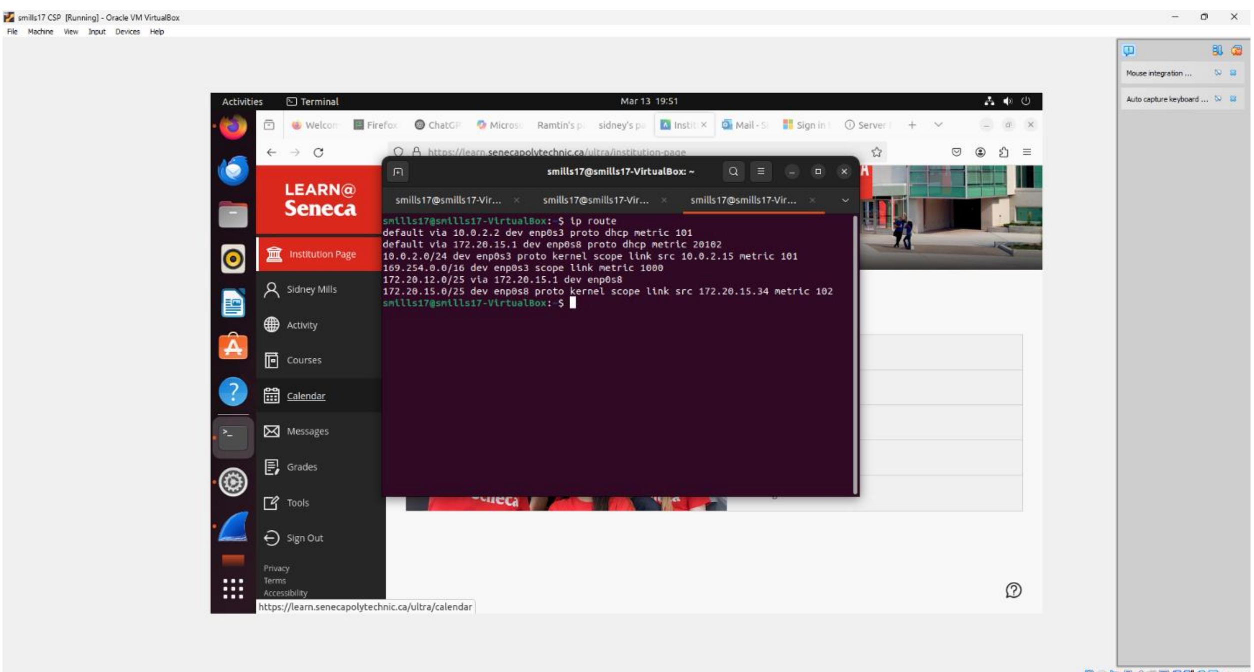
ramtin@ramtin-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:e8:f1:40 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 78680sec preferred_lft 78680sec
inet6 fe80::a00:27ff:fee8:f140/64 scope link
valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:c1:9d:f2 brd ff:ff:ff:ff:ff:ff
inet 10.10.10.99/28 brd 10.10.10.111 scope global noprefixroute enp0s8
valid_lft forever preferred_lft forever
inet6 fe80::a2ec:9df2:d64f:9e62/64 scope link noprefixroute
valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:23:a4:fe brd ff:ff:ff:ff:ff:ff
inet 172.20.12.79/25 brd 172.20.12.127 scope global dynamic noprefixroute enp0s9
valid_lft 3072sec preferred_lft 3072sec
inet6 fe80::d637:bdab:8305:a768/64 scope link noprefixroute
valid_lft forever preferred_lft forever
ramtin@ramtin-VirtualBox:~$



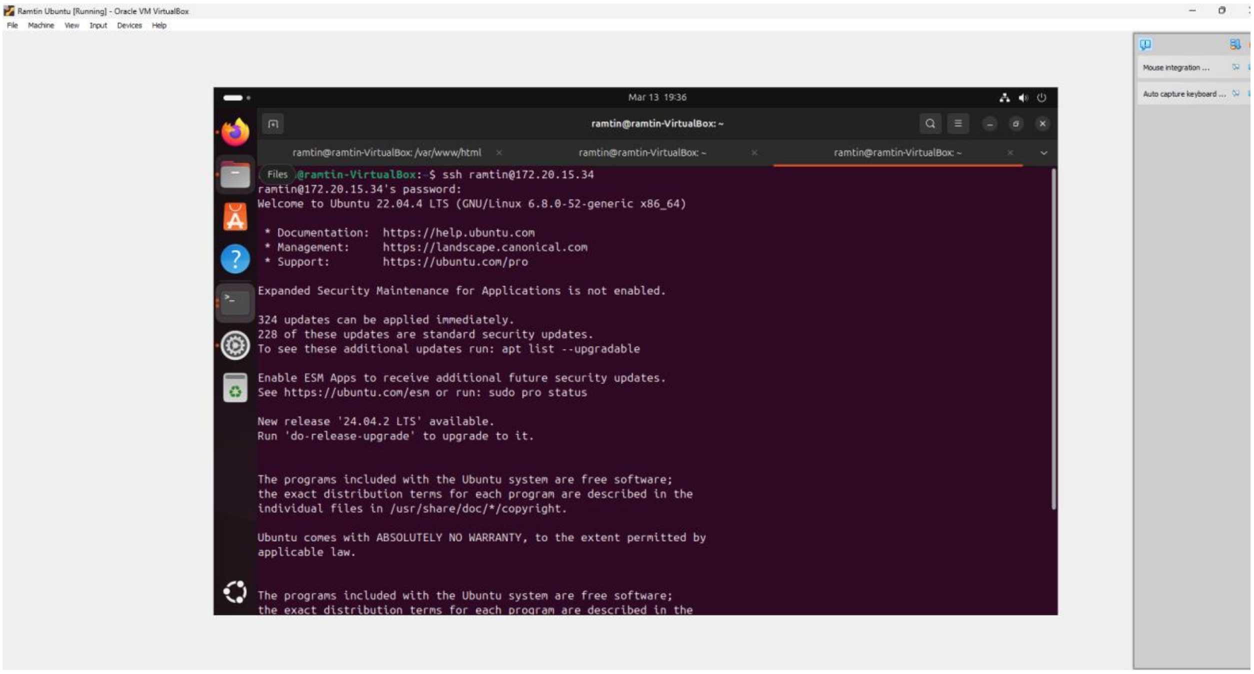
smills17@smills17-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:5c:d0:c3 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 82064sec preferred_lft 82064sec
inet6 fe80::34a1:abd4:bb5f:776b/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:09:6d:9f brd ff:ff:ff:ff:ff:ff
inet 172.20.15.34/25 brd 172.20.15.127 scope global dynamic noprefixroute enp0s8
valid_lft 2526sec preferred_lft 2526sec
inet6 fe80::12ce:894d:bc73:ed3a/64 scope link noprefixroute
valid_lft forever preferred_lft forever
smills17@smills17-VirtualBox:~$

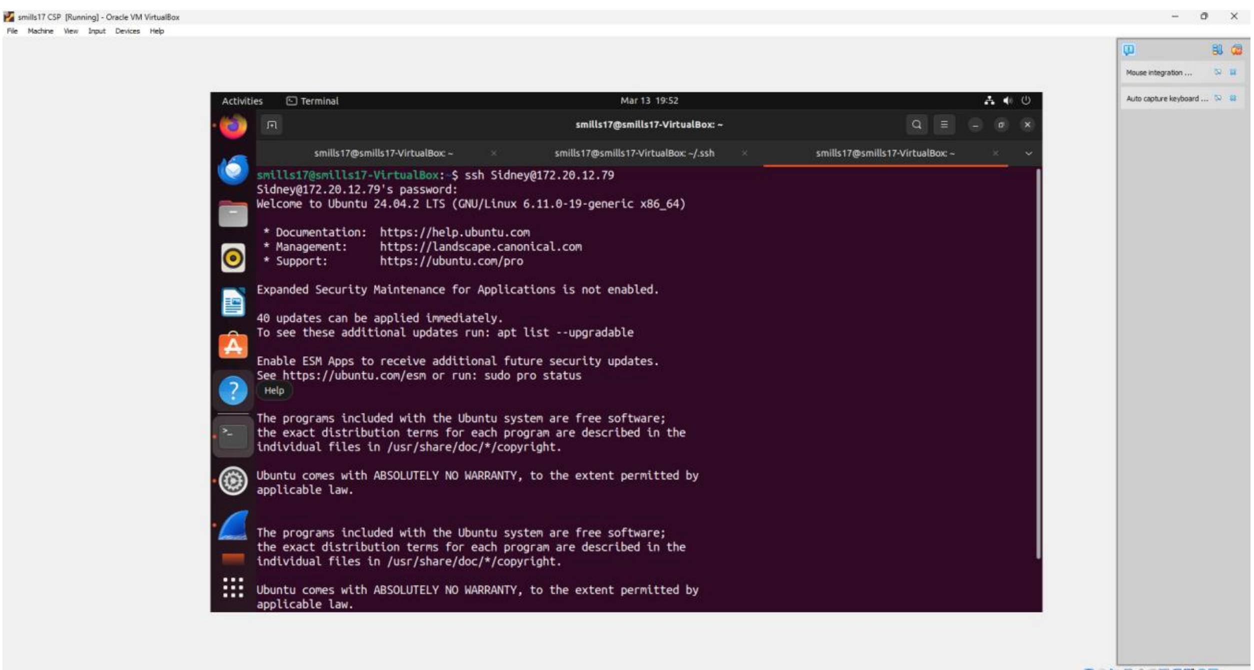
4. Ip route


Ramtin Ubuntu [Running] - Oracle VM VirtualBox
Mar 13 19:35
ramtin@ramtin-VirtualBox: ~
ramtin@ramtin-VirtualBox: /var/www/html
ramtin@ramtin-VirtualBox: ~
ramtin@ramtin-VirtualBox:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
default via 172.20.12.1 dev enp0s9 proto dhcp src 172.20.12.79 metric 20104
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.10.10.96/28 dev enp0s8 proto kernel scope link src 10.10.10.99 metric 103
172.20.12.0/25 dev enp0s9 proto kernel scope link src 172.20.12.79 metric 104
172.20.15.0/25 via 172.20.12.1 dev enp0s9
ramtin@ramtin-VirtualBox:~$

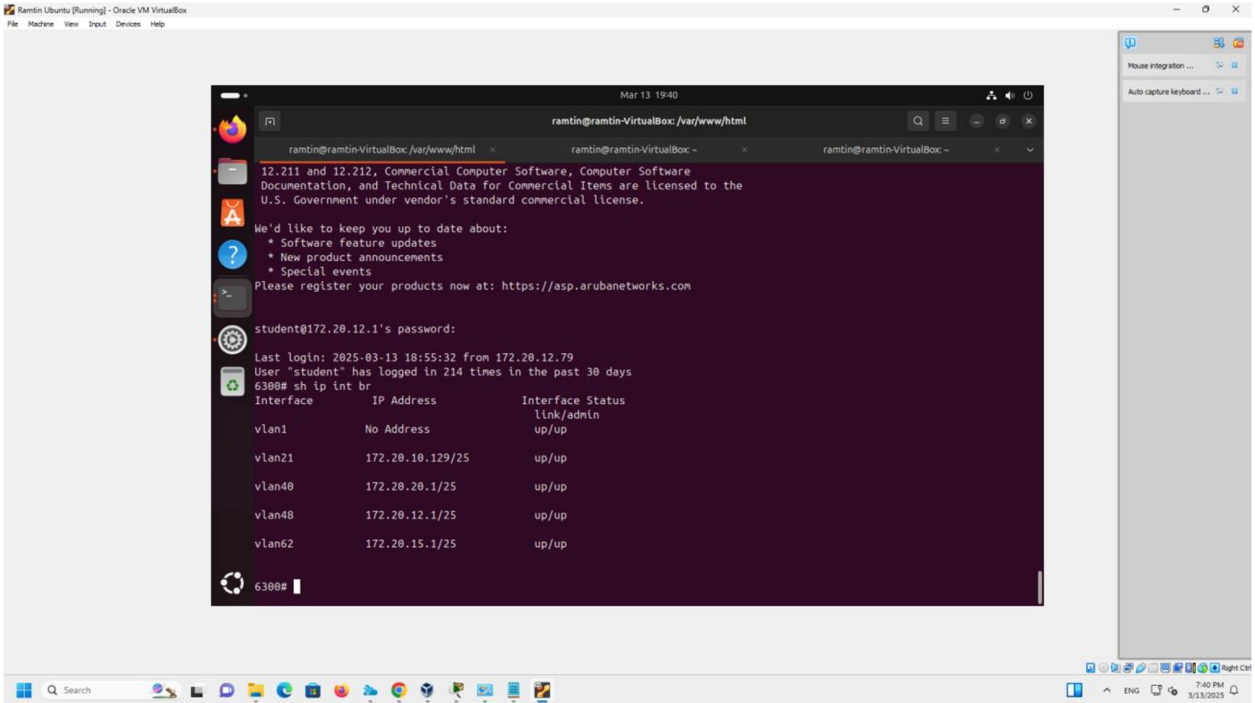

smills17 CSP [Running] - Oracle VM VirtualBox
Mar 13 19:51
smills17@smills17-VirtualBox: ~
smills17@smills17-VirtualBox:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp metric 101
default via 172.20.15.1 dev enp0s8 proto dhcp metric 20102
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 101
169.254.0.0/16 dev enp0s3 scope link metric 1000
172.20.12.0/25 via 172.20.15.1 dev enp0s8
172.20.15.0/25 dev enp0s8 proto kernel scope link src 172.20.15.34 metric 102
smills17@smills17-VirtualBox:~$
LEARN@ Seneca
Institution Page
Sidney Mills
Activity
Courses
Calendar
Messages
Grades
Tools
Sign Out

5. Successfully SSH'ed to other VM

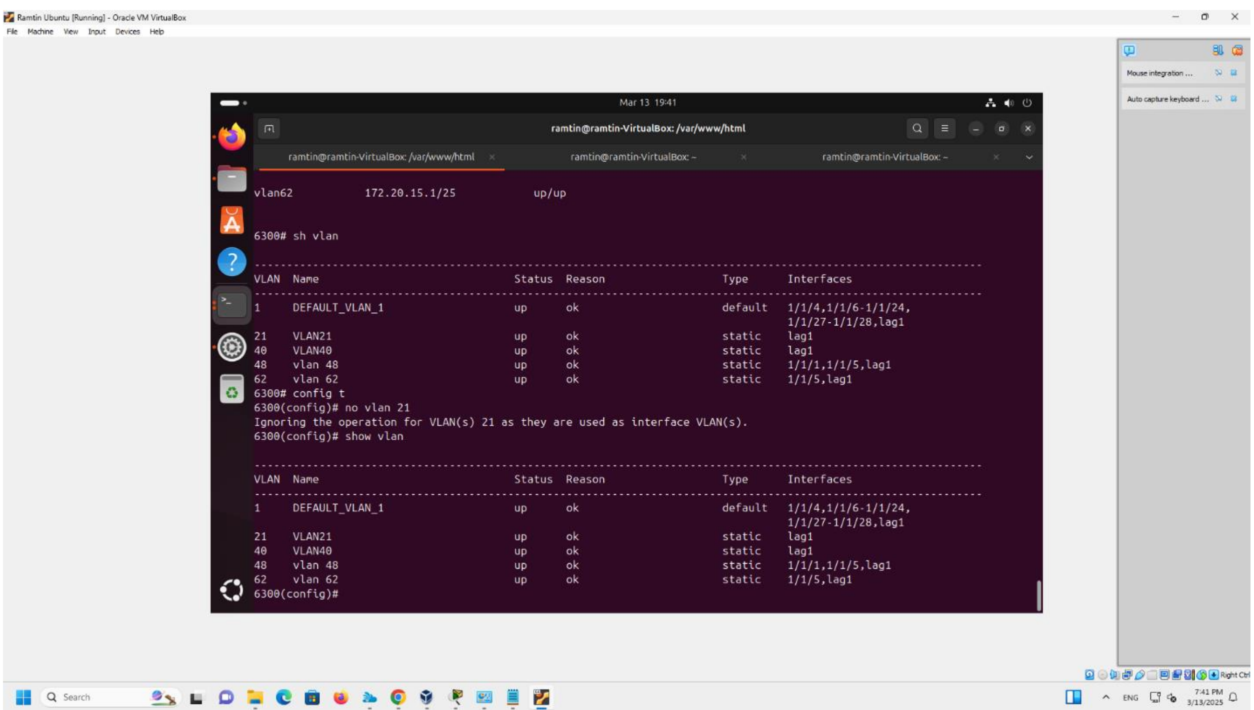


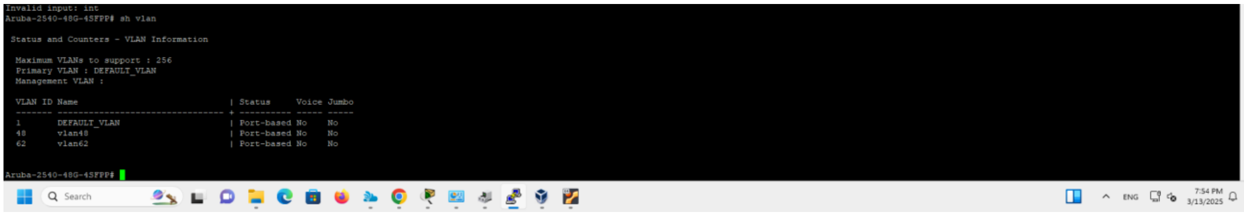


6. sh ip int br

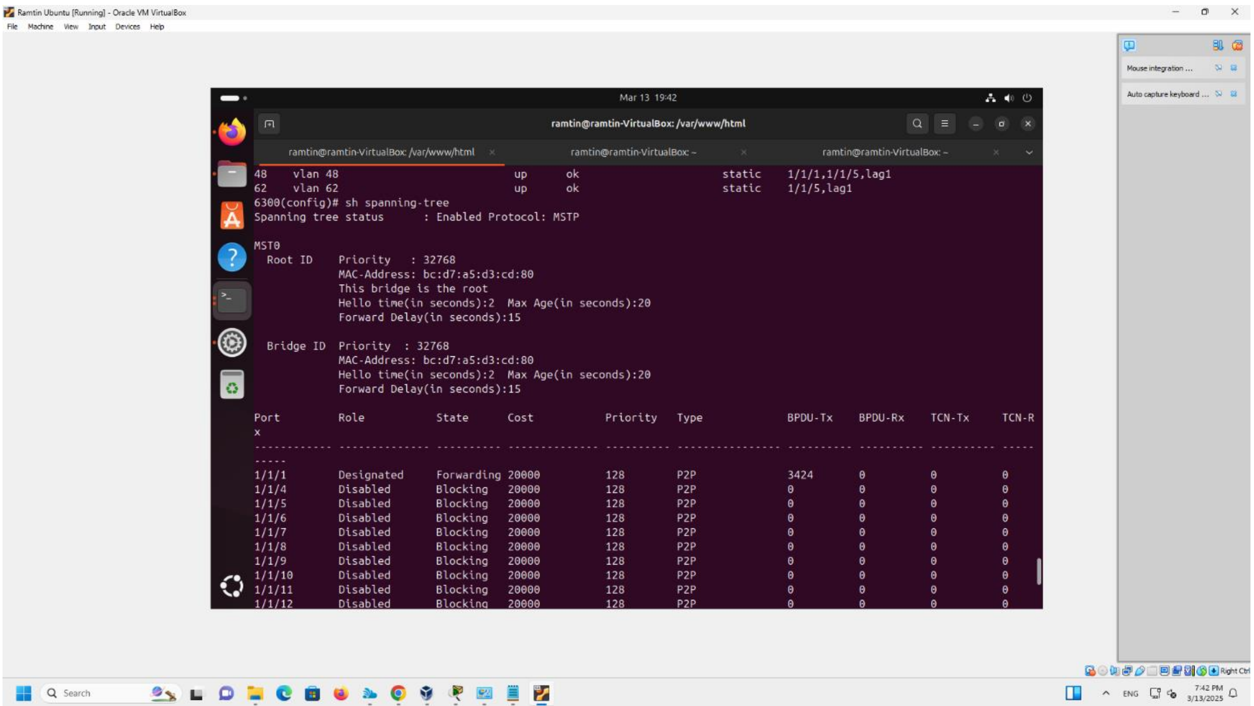


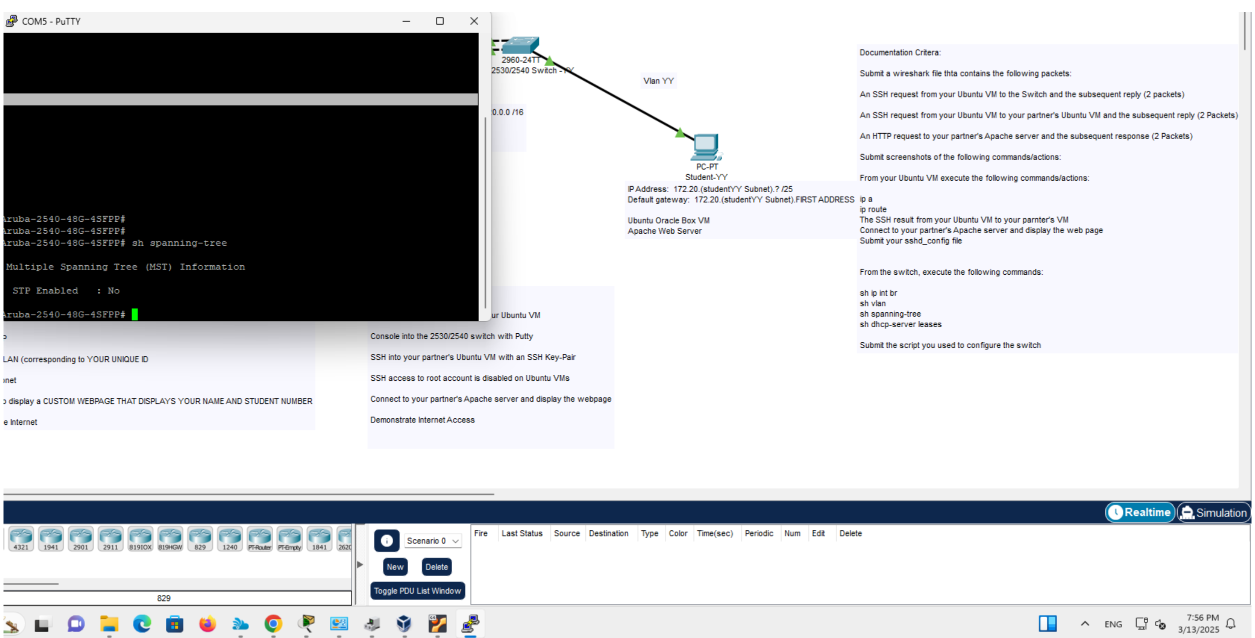
7. show vlan



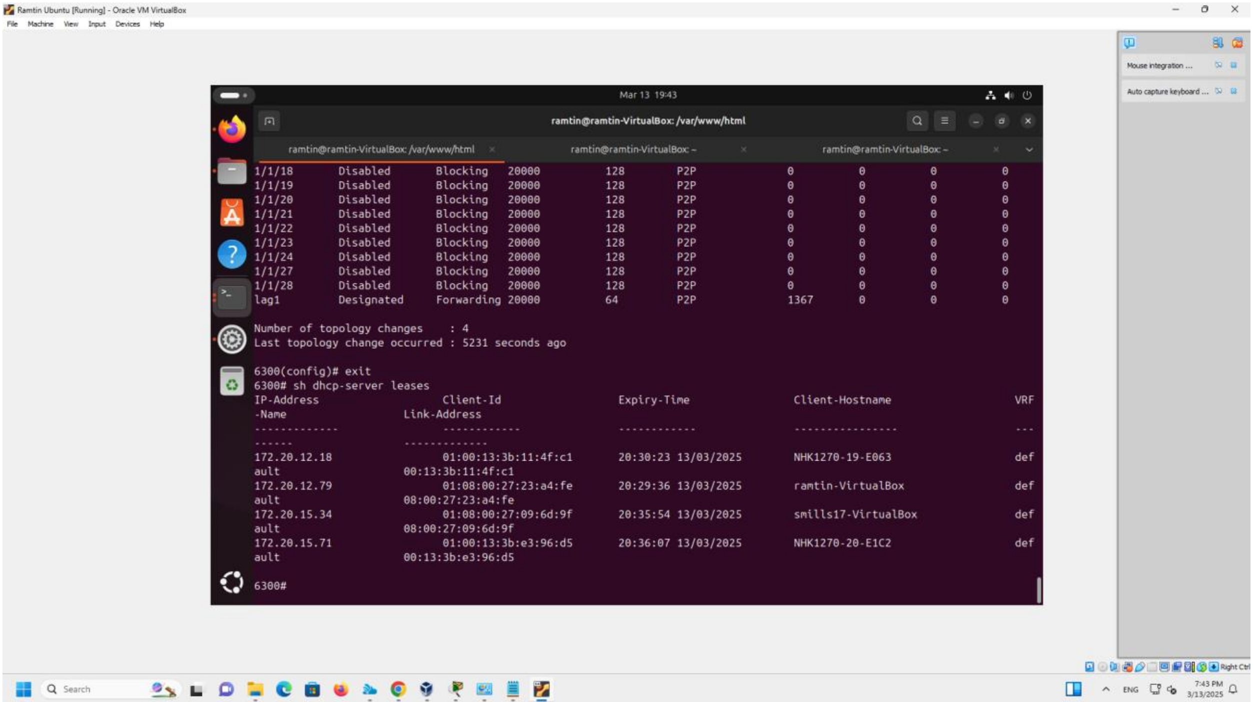


8. show spanning-tree

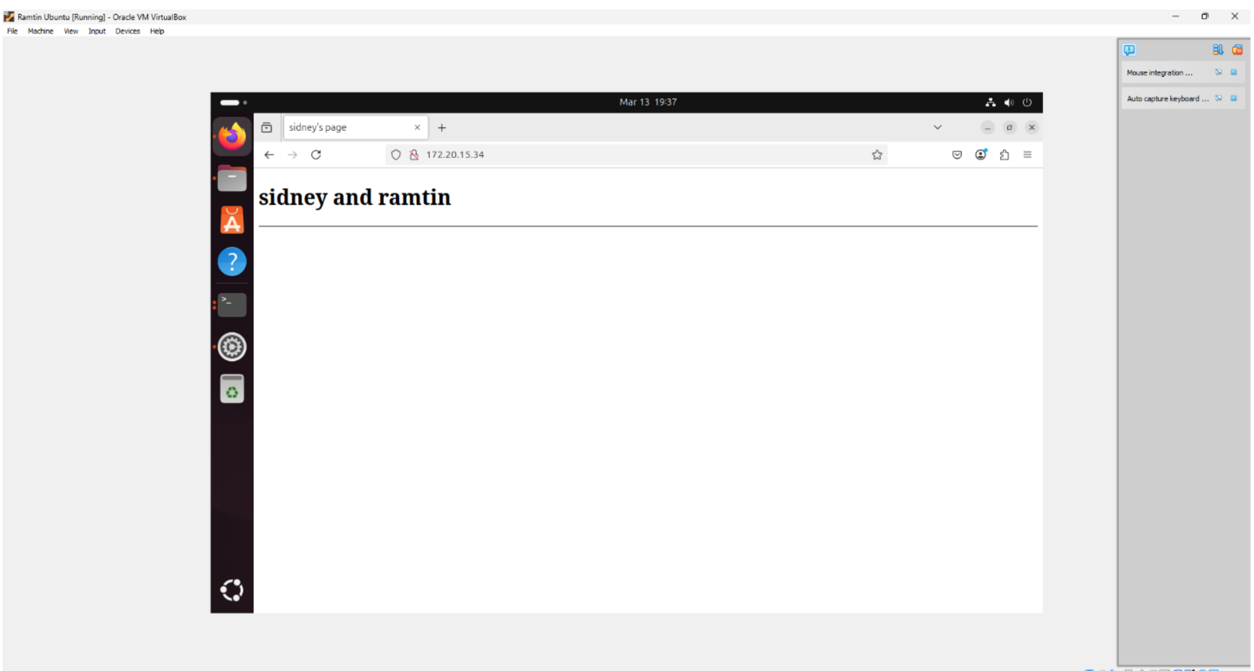


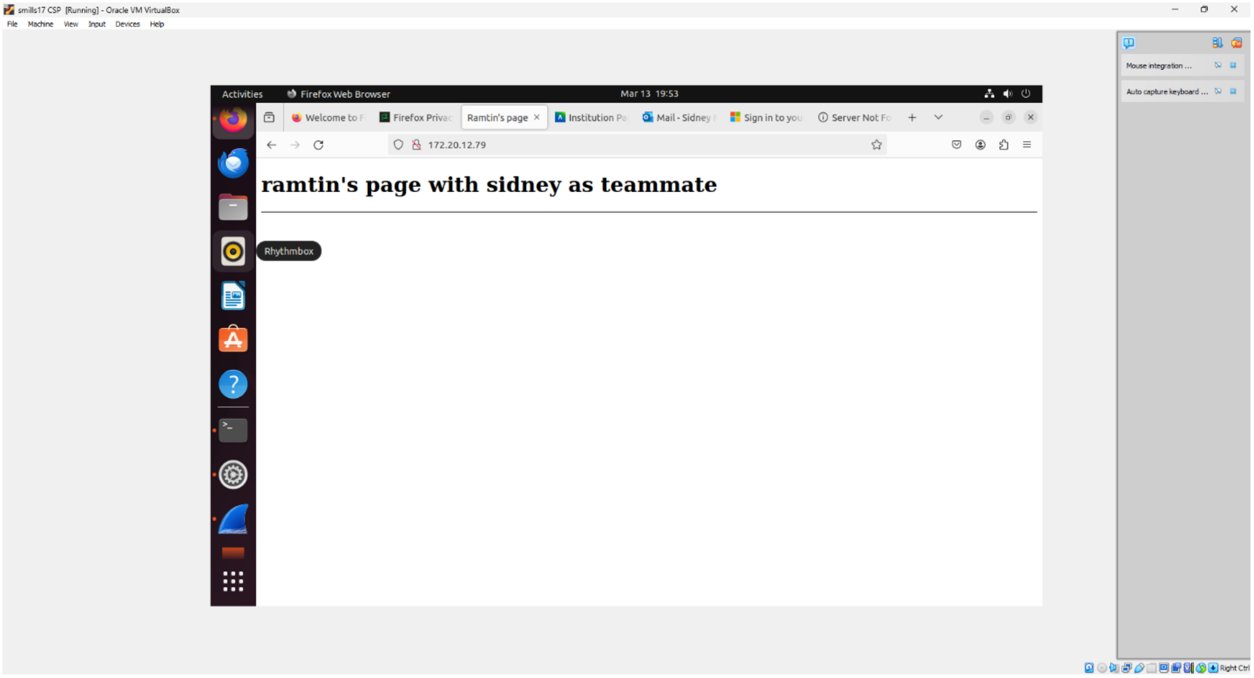


9. sh dhcp-server leases

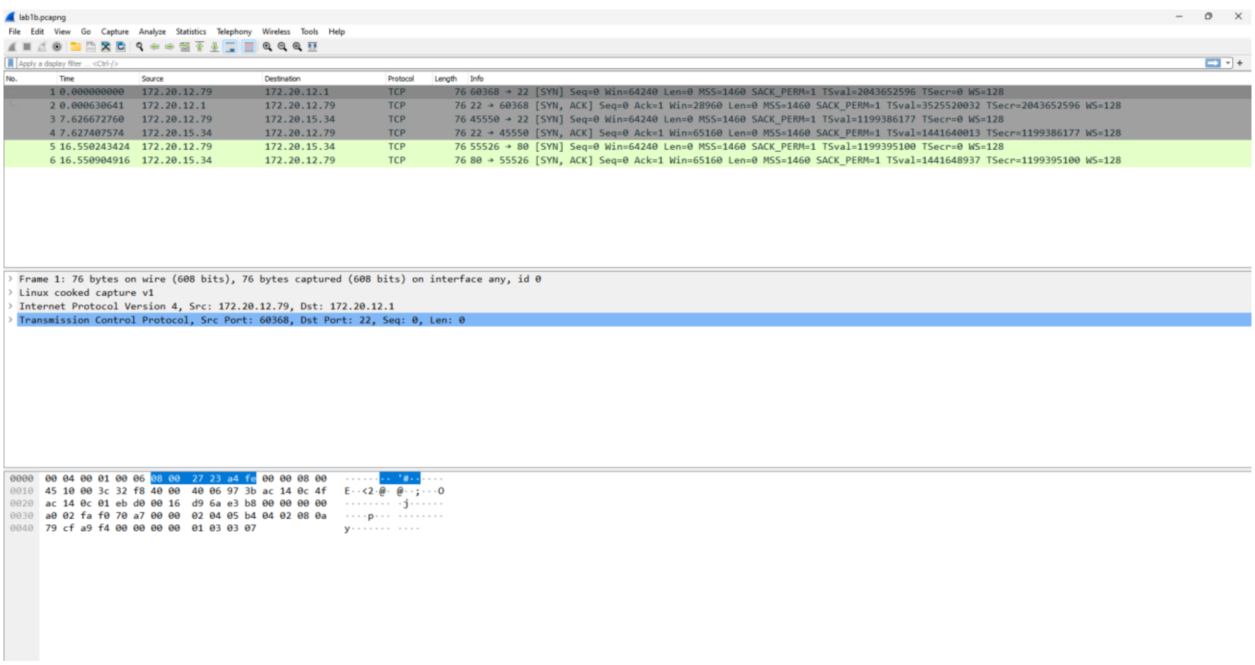


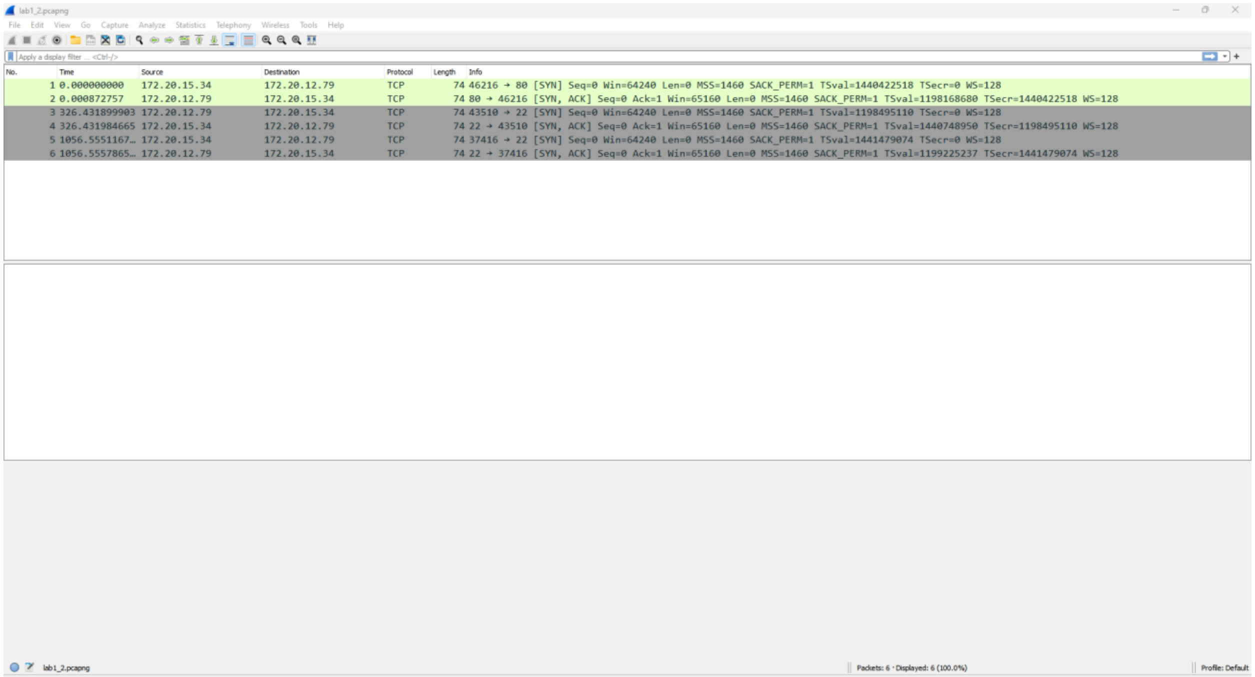
10. webpage





11. wireshark capture





lab1_2.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter … <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 172.20.15.34 172.20.12.79 TCP 74 46216 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1440422518 TSecr=0 WS=128
2 0.000872757 172.20.12.79 172.20.15.34 TCP 74 80 → 46216 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1198168680 TSecr=1440422518 WS=128
3 326.431899903 172.20.12.79 172.20.15.34 TCP 74 43510 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1198495110 TSecr=0 WS=128
4 326.431984665 172.20.15.34 172.20.12.79 TCP 74 22 → 43510 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1440748950 TSecr=1198495110 WS=128
5 1056.5551167… 172.20.15.34 172.20.12.79 TCP 74 37416 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1441479074 TSecr=0 WS=128
6 1056.5557865… 172.20.12.79 172.20.15.34 TCP 74 22 → 37416 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1199225237 TSecr=1441479074 WS=128
lab1_2.pcapng
Packets: 6 · Displayed: 6 (100.0%)
Profile: Default